

Sri Lanka Institute of Information Technology

Data Warehousing and Business Intelligence

Assignment 1 – 2025, Semester

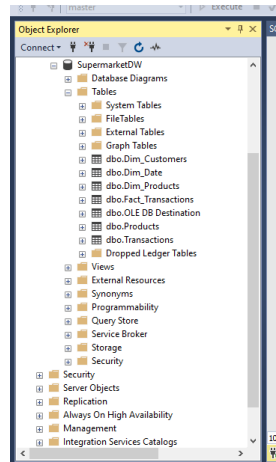
Documentation



S.SADURSHAN IT22180520

Data source

The date The Supermarket_DW tables, which were used for this project, were replicated from assignment Dimensional and fact tables exist in this data Dim_Date, Dim_Products, Dim_Customers, and Fact_Transactions. These tables, holding a year's worth of simulated retail transaction data available for OLAP analysis, are implemented using a Star Schema design.



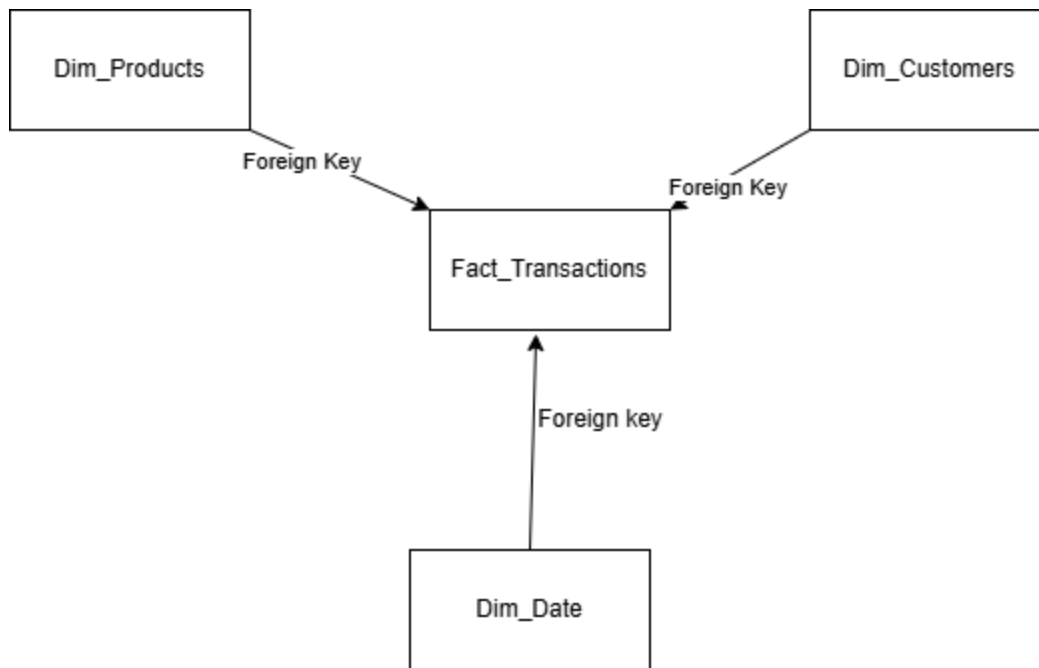
use SupermarketDW;
SELECT TOP 10 * FROM Fact_Transactions;

TransactionID	CustomerKey	ProductKey	DateKey	Quantity	acorn_txn_create_time	acorn_txn_complete_time	txn_process_time_hours
1	31	11	20240905	7	2025-04-18 16:28:12.910	2024-01-01 11:33:00.000	-11357
2	62	14	20240131	7	2025-04-18 16:28:12.910	2024-01-01 09:00:00.000	-11359
3	22	25	20240718	1	2025-04-18 16:28:12.910	2024-01-01 10:12:00.000	-11358
4	34	30	20240804	5	2025-04-18 16:28:12.910	2024-01-01 10:45:00.000	-11358
5	90	47	20241204	9	2025-04-18 16:28:12.910	2024-01-01 09:38:00.000	-11359
6	92	32	20240407	3	2025-04-18 16:28:12.910	2024-01-01 09:30:00.000	-11359
7	38	14	20241023	1	2025-04-18 16:28:12.910	2024-01-01 10:25:00.000	-11358
8	95	35	20240609	1	2025-04-18 16:28:12.910	2024-01-01 13:07:00.000	-11355
9	8	4	20240901	10	2025-04-18 16:28:12.910	2024-01-01 13:15:00.000	-11355
10	65	34	20240130	3	2025-04-18 16:28:12.910	2024-01-01 13:02:00.000	-11355

Query executed successfully.

DESKTOP-867T74T (16.0 RTM) | DESKTOP-867T74T.Buwane... | SupermarketDW | 00:00:00 | 10 rows

Ln 2 Col 1 Ch 1 INS



Star Schema

The database was created and popularized using SSIS ETL pipelines and present in Microsoft SQL Server and used as sources for both SQL Server Analysis Services (SSAS) and Power BI.

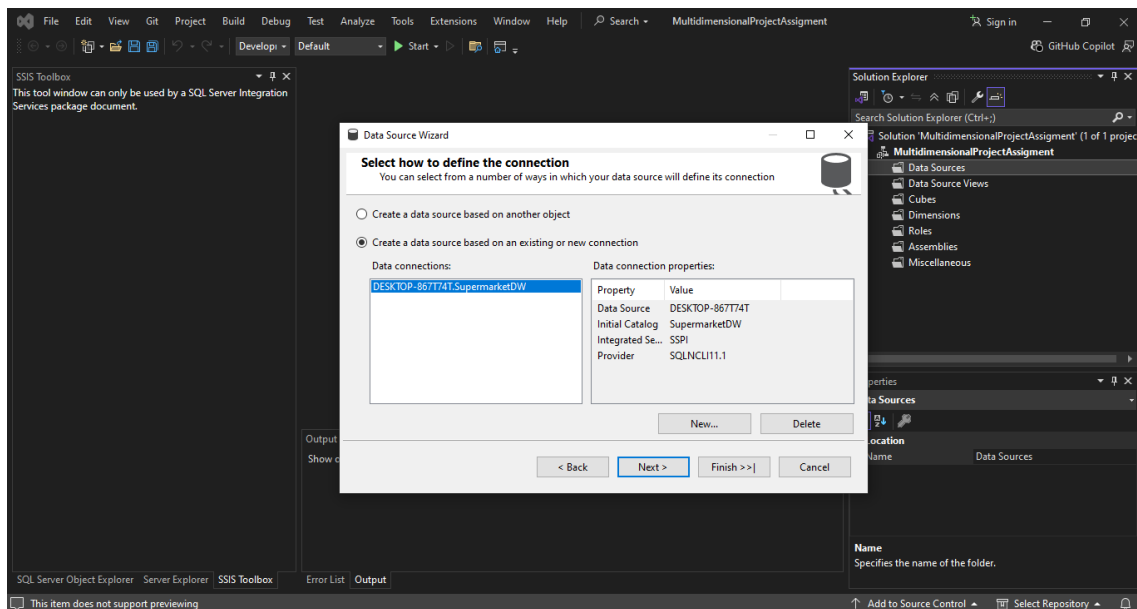
SSAS Cube Implementation

The following dimensions and SQL Server Data tools were used to develop the SupermarketDWCube cube in SQL Server Analysis Services. Dim_Customer

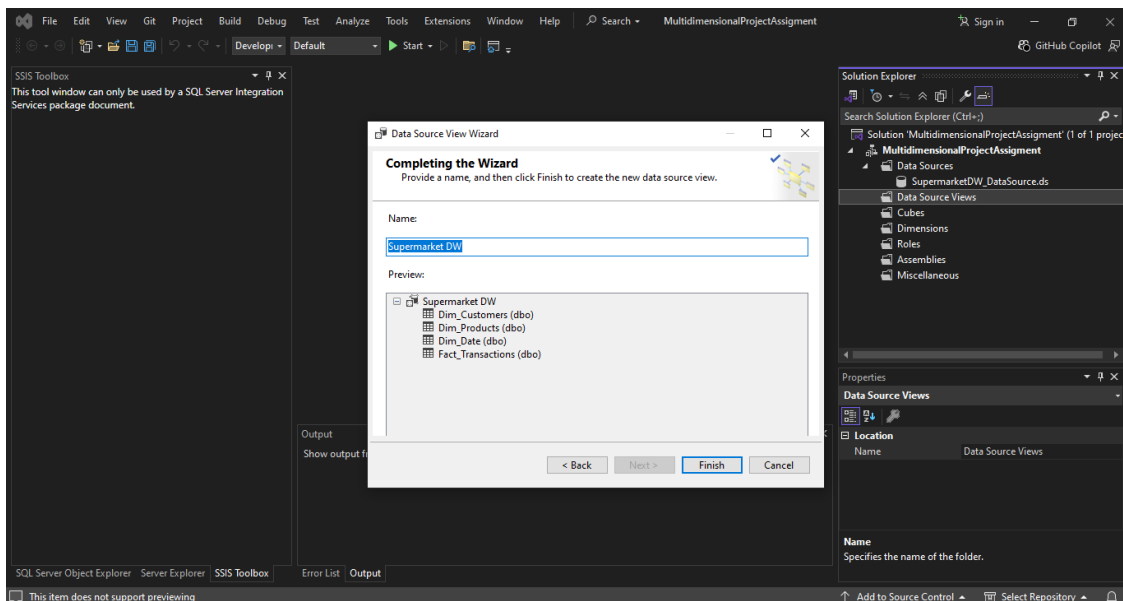
- Dim_Product
- Dim_Date

And the keys that used to include are

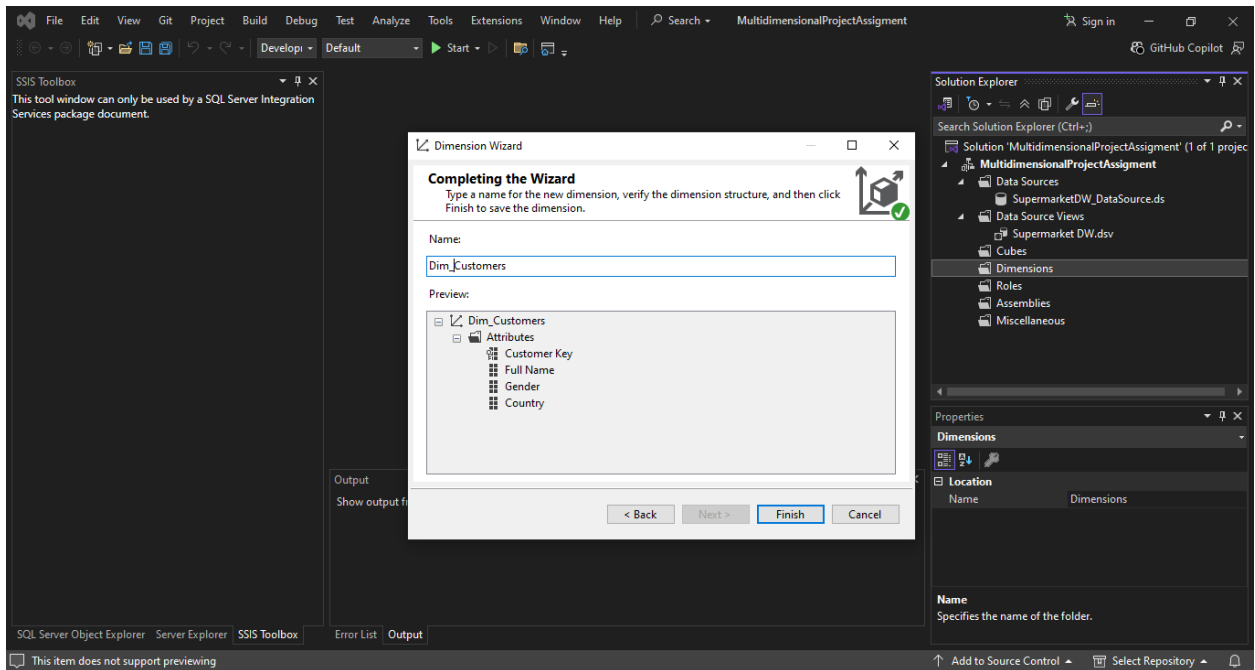
- Quantity in Dim_product
- txn_process_time_hours in dim_date



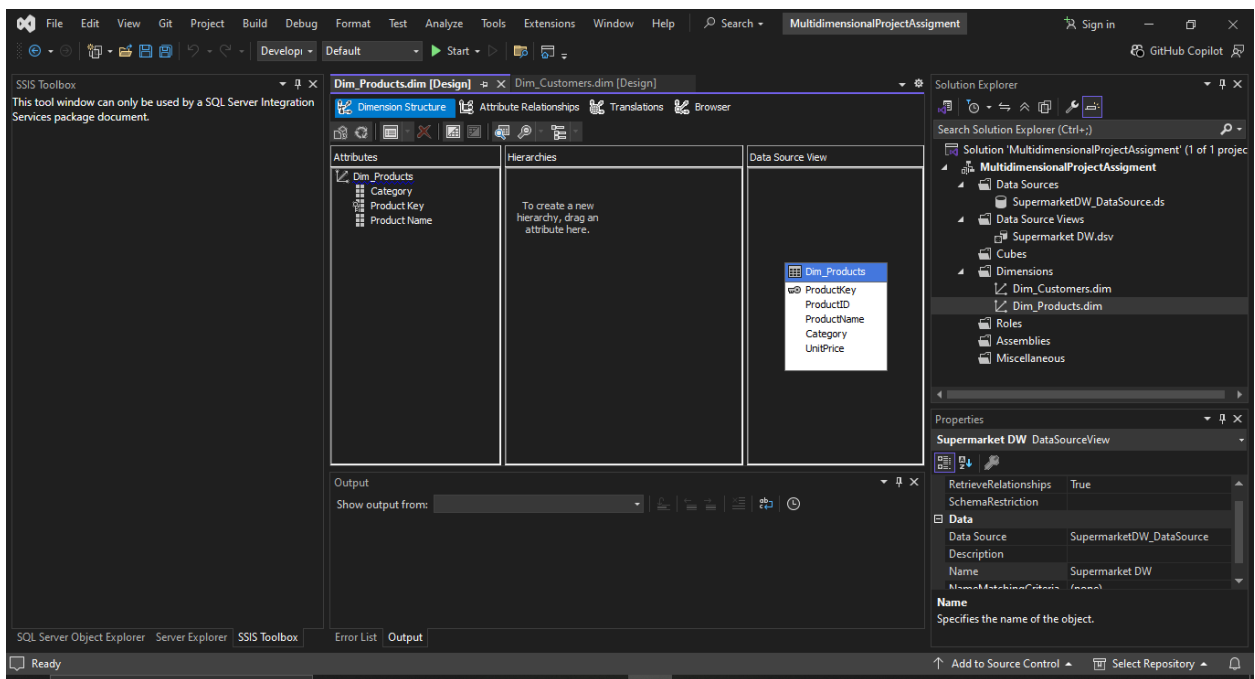
Connection Establishment To Data-Source



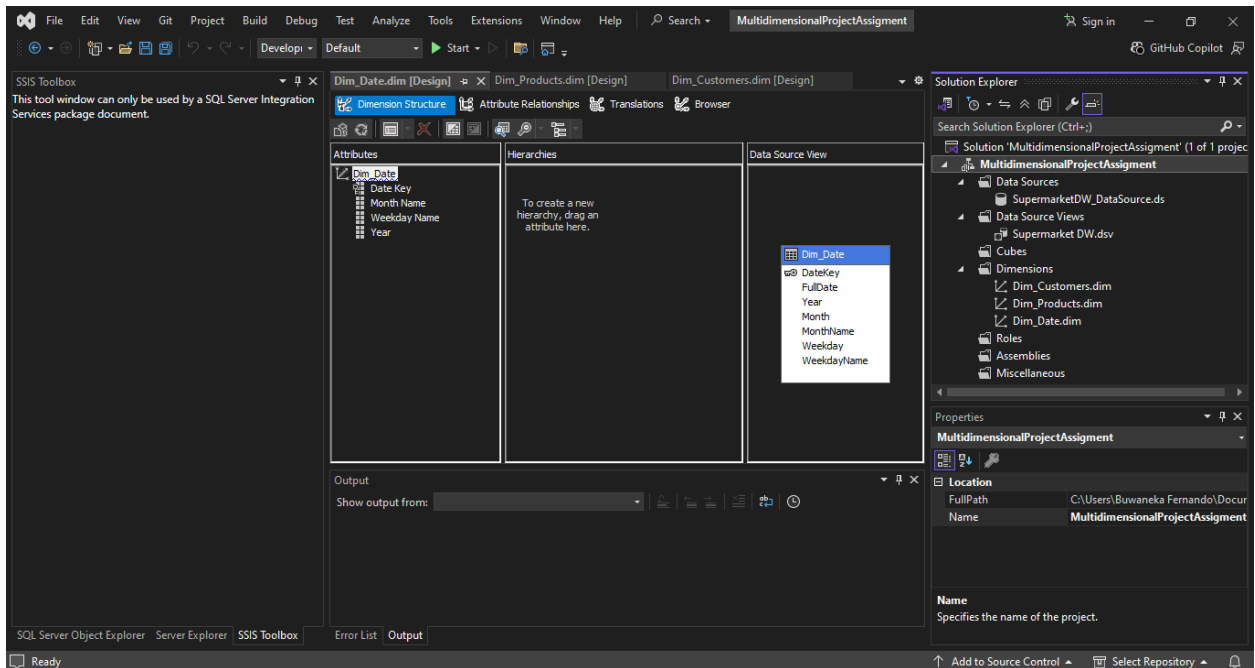
Display Data-sources



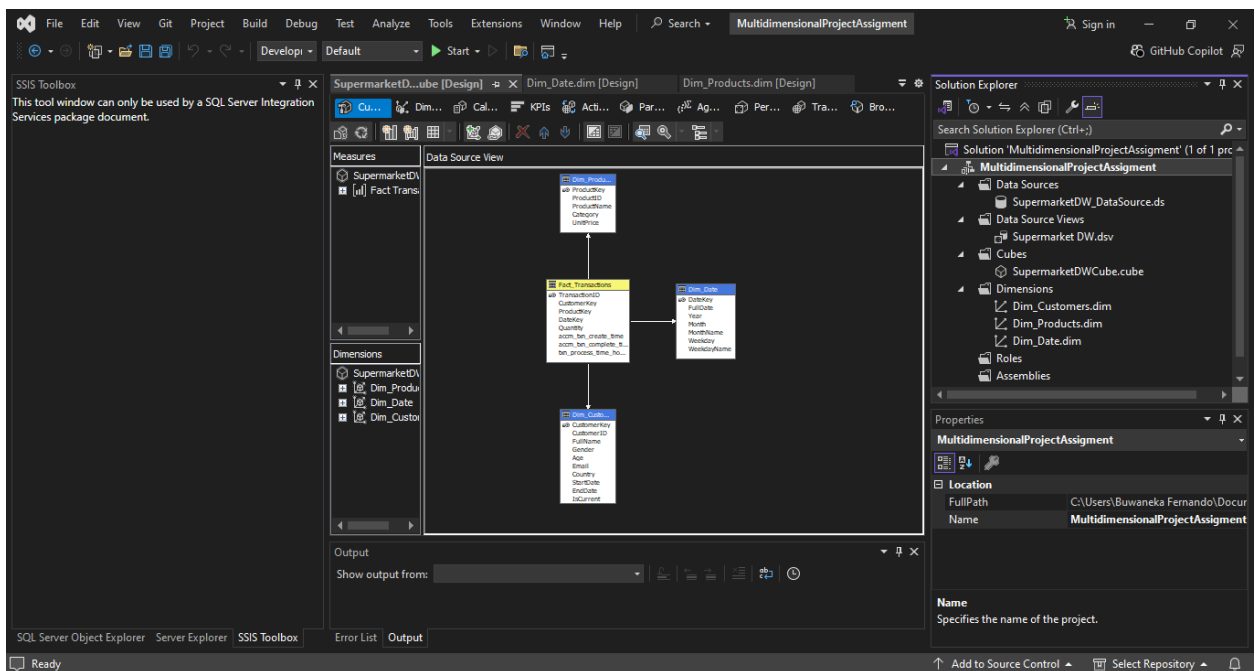
Selecting Customer attributes



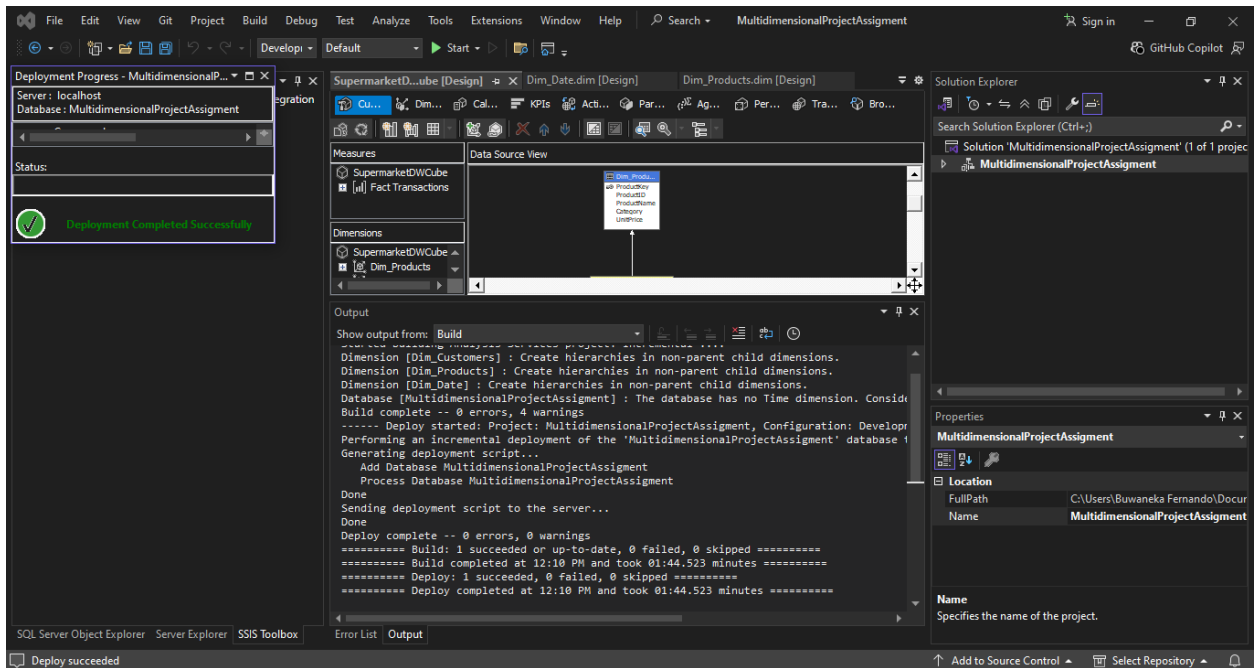
Selecting Product attributes



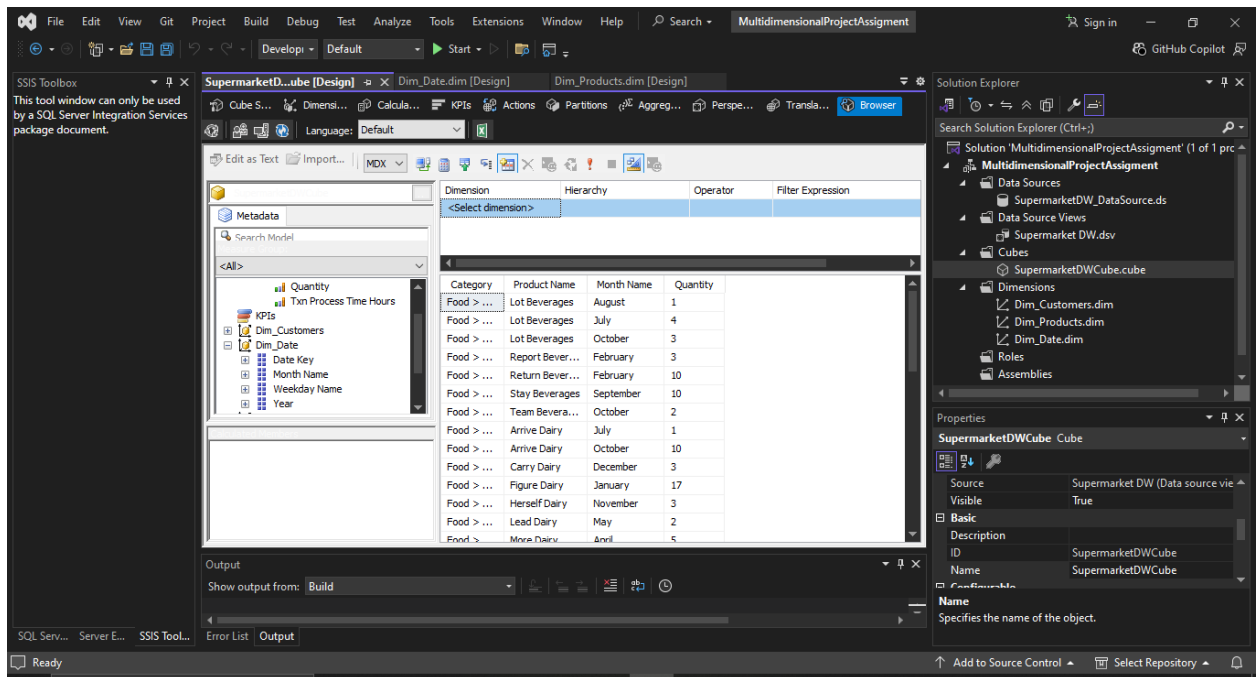
Selecting Date Dimension



Dimension and measuring



Cube



Excel OLAP Operations

Excel was connected to the SSAS cube using the Analysis Services connector. In Excel Sheet, PivotTable was created to analyze the quantity sold by product category across months.

The screenshot displays the Microsoft Excel interface. The main window shows a PivotTable with the following data:

Quantity	Column Labels	August	December	February	January	July	June	March	May	November	October	September	Grand Total
Food > Beverages	April	1		13		4			2		3	10	33
Food > Dairy		5	5	3		17	2						47
Food > Snacks			3		1	7		8	2	9	5	9	51
Home Care > Cleaning		3		1	2	3		1				4	14
Home Care > Laundry		6		9					6		5	10	41
Grand Total		14	9	13	16	27	6	9	2	17	13	38	186

The PivotTable Fields task pane on the right shows the following configuration:

- Dim_Date** is selected in the list.
- Month Name** is checked under the Dim_Date list.
- Filters:** (Empty)
- Columns:** Month Name
- Rows:** Category
- Values:** Quantity

The status bar at the bottom indicates "Ready" and "Accessibility: Good to go".

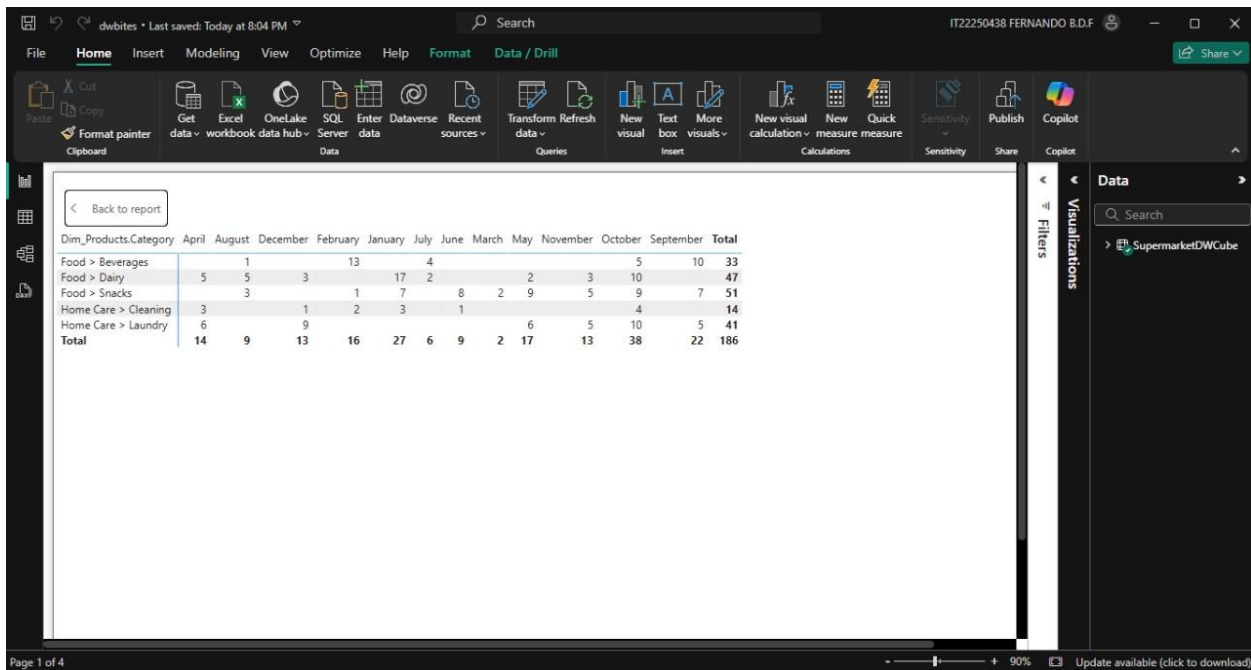
Power BI Reports

Report 1: Matrix

Values of Product's Quantity by Category, which are rows, and Date's MonthName, which are columns, are displayed in this report using a Power BI matrix chart.

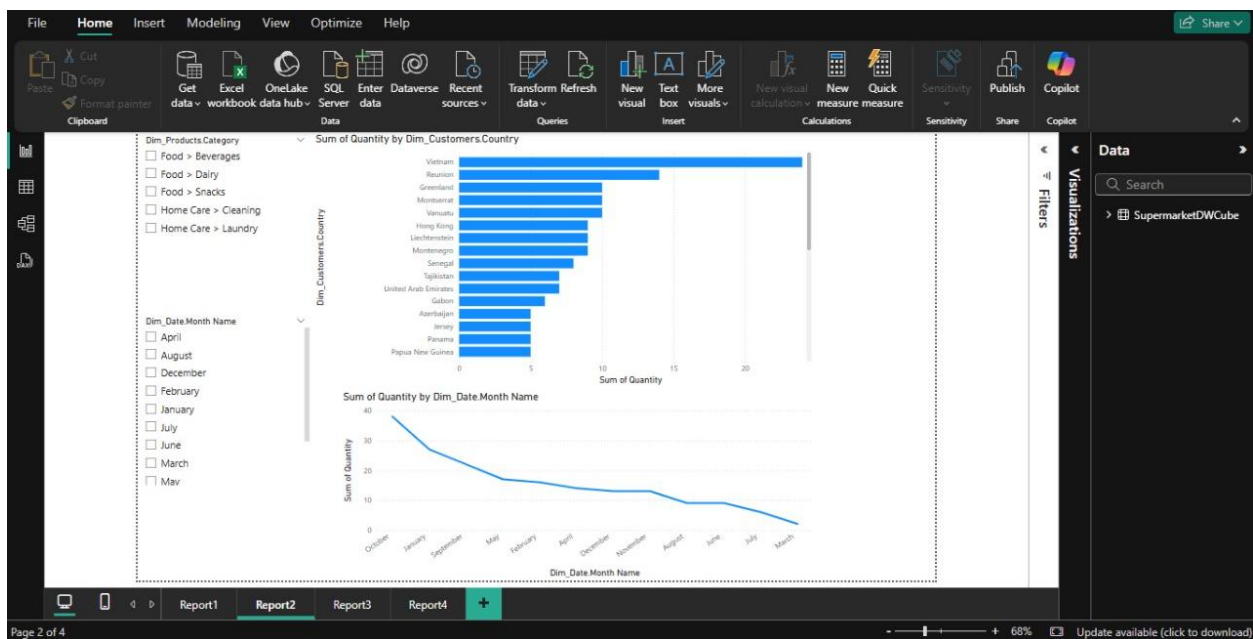
Steps :

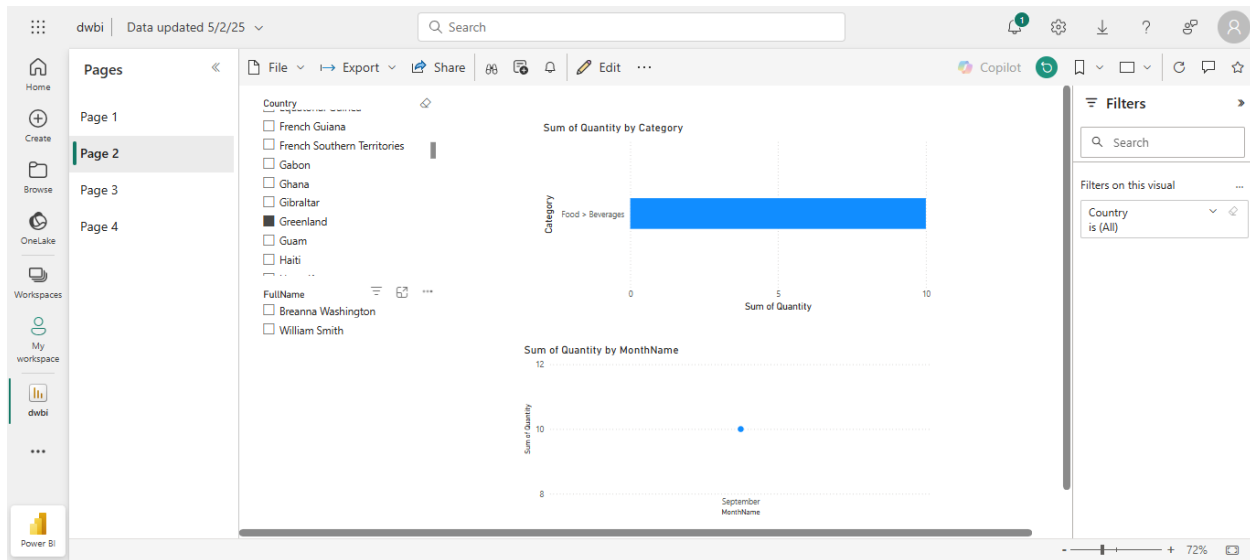
- Open page and drag Matrix Icon from Visualization
- Dragging the fields from dim tables which Category from DimProduct to Rows , Month Name from Dim_Date to Column ,and Quantity from Fact_Transaction to Values



Report 2: Charts and cascade sliding

For interactive control by the users, the Country slicer behaves as a filter on the data available in the Customer Name slicer. For analysis of product categories and time by quantity in trends for a given month, two interactive visualizations were done: a bar chart and a line chat.



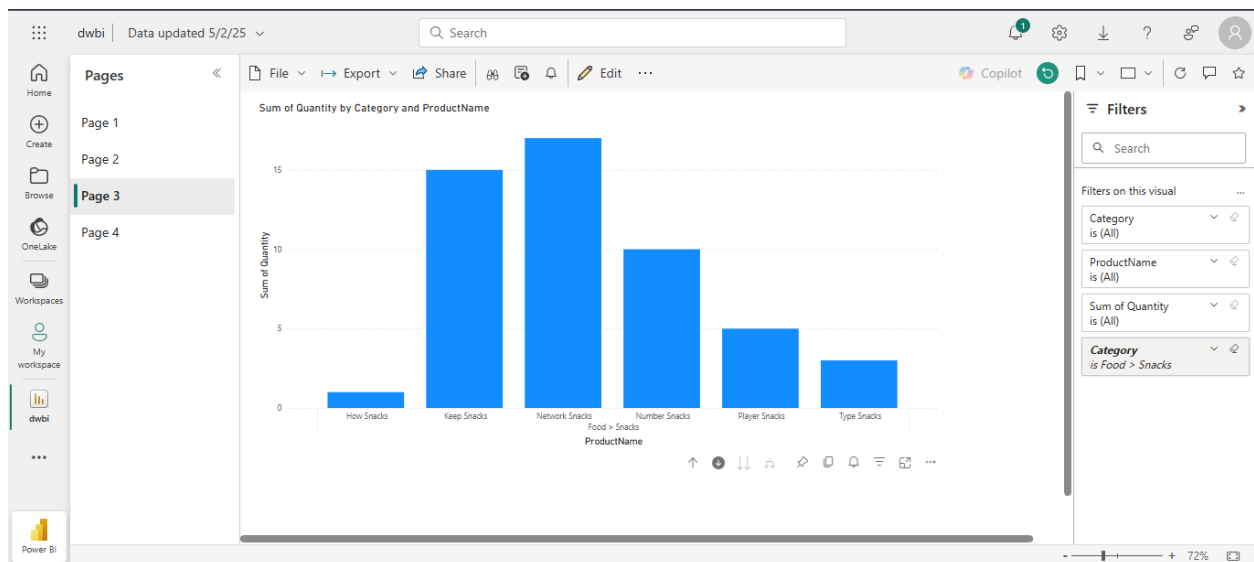
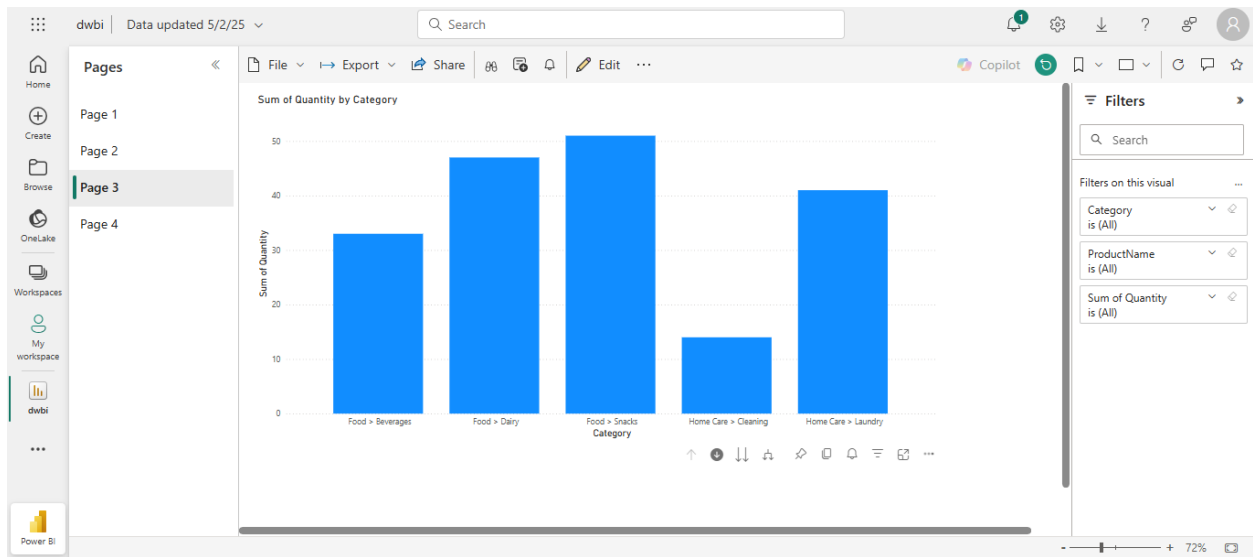


Steps:

- Open page and drag two slicer for country and Full name for the customer
- Drag Two charts and include related fields
- Then the chart will present the data according to the slicer selection or whole data

Report 3: Drill-Down

In this report, it used a drill-down from Product Category to Product Name using a column chart. This helps users to explore how product categories break down into individual item with their Quantity.

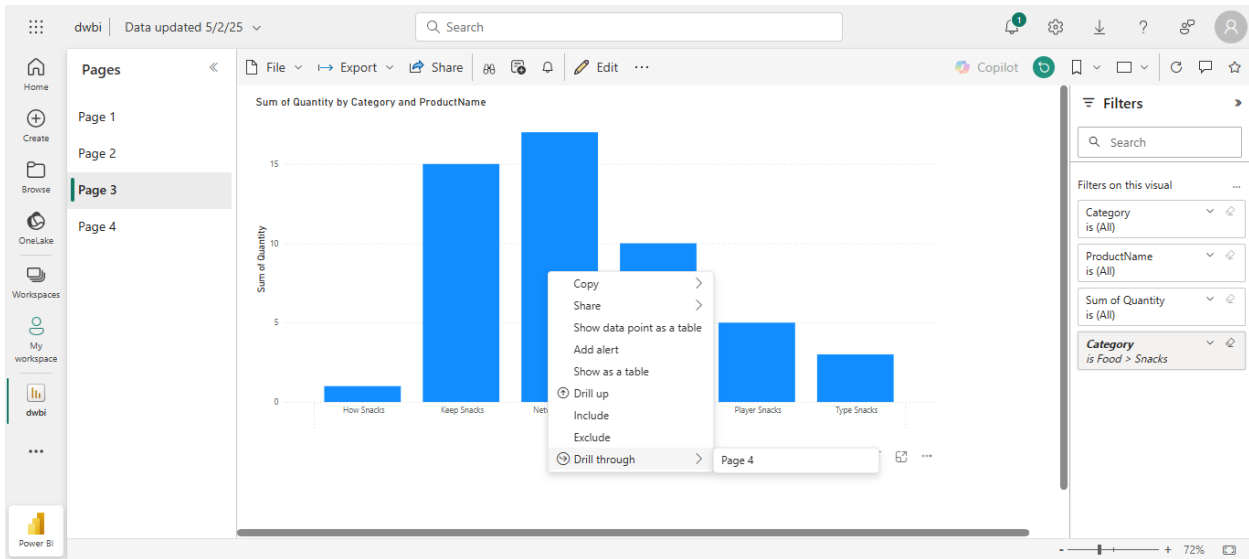


Steps:

- Drag a chart
- Add the category and the product name to the x axis
- drag Quantity into Values
- Enable Drill-Down

Report 4: Drill-Through

employs a drill-through page filter using Product Category. Related transactional information is shown when you right-click on a category in the main report and go to the detail page



The screenshot shows the Microsoft Power BI Desktop interface. The main area displays a table with the following data:

Dim_Products.Product Name	Sum of Quantity	Sum of Txn Process Time Hours	Dim_Customers.Full Name
Network Snacks	7	-11359	Alexis Gonzalez
Network Snacks	9	-11355	Jennifer Cummings
Network Snacks	1	-11358	Jonathan Moody
Total	17	-34072	

The interface includes a top ribbon with tabs like File, Home, Insert, Modeling, View, Optimize, and Help. The right-hand pane shows the 'Data' section with a search bar and a list of data sources, including 'SupermarketDWCube'. The bottom status bar indicates 'Page 4 of 4' and a zoom level of 120%.

Steps is Explained below :

Drag a field, add a table, and fill it in with the following information: Product name, quantity, date, and customer name .Returning to the report 3 chart page
To view the specifics of the pertinent filtered detail, right-click and select Drill-Through.

Conclusion

This exercise uses the data warehouse constructed in Assignment 1 to demonstrate and take advantage of the entire OLAP lifecycle.

It covered an interactive report with PowerBI, OLAP analysis with Excel, and a cude build with SSAS.